

MATHEMATICS SUBJECTIVE TEST

Class X (Session 2026 – 27)

Duration : 120 Minutes

Max. Marks: 45

General Instruction

1. All questions are compulsory.
2. Section A – 5 questions $\times$ 1 mark = 5 marks.
3. Section B – 5 questions $\times$ 3 marks = 15 marks.
4. Section C – 5 questions $\times$ 5 marks = 25 marks.

SECTION – A

(Each question carries 1 mark)

1. Without calculating, determine whether $(\sqrt{5} + \sqrt{3})^2 + (\sqrt{5} - \sqrt{3})^2$ is rational or irrational. Justify.
2. If one zero of $(a^2 + 4)x^2 + 13x + 4a$ is the reciprocal of the other, find a.
3. Find the value of k for which $2x + ky = 1$ and $5x + (k + 4)y = 2$ has no solution.
4. If $x = -2$ is a root of $3x^2 + 7x + p = 0$, find the other root without solving.
5. LCM of two numbers is 14 times their HCF. Sum of LCM and HCF = 600. One number is 280. Find the other.

SECTION - B

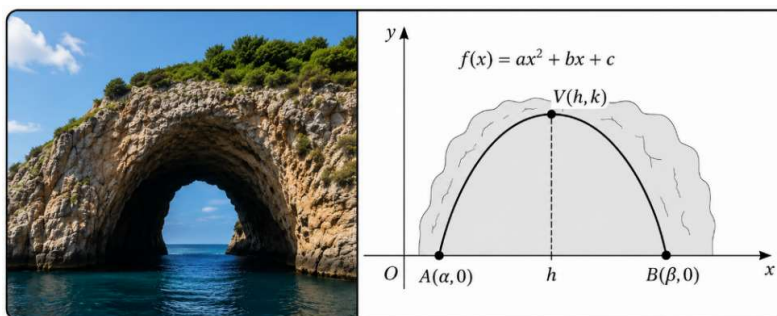
(Each question carries 3 mark)

6. Prove that $1/(2 + \sqrt{3})$ is irrational.
7. α and β are zeroes of $3x^2 - 7x + 2$. Form a polynomial whose zeroes are $(2\alpha + 3\beta)$ and $(3\alpha + 2\beta)$.
8. Two taxi companies offer the following fare structures : Company A charges ₹60 as a fixed charge for the first kilometre and ₹10 per kilometre for every subsequent kilometre. Company B charges ₹40 as a fixed charge for the first kilometre and ₹15 per kilometre for every subsequent kilometre. Find the distance at which both companies charge the same fare. Also determine for what distances Company B is cheaper than Company A.
9. Find roots of $3\sqrt{3}x^2 + 10x + \sqrt{3} = 0$ by splitting the middle term.
10. α and β are zeroes of $2x^2 - 5x + k$. Find k given $\alpha^2 + \beta^2 + \alpha\beta = 21/4$.

SECTION - C

(Each question carries 5 mark)

11. A group of school friends went on an expedition to see caves. One person remarked that the entrance of the cave resembles a parabola and can be represented by a quadratic polynomial $f(x) = ax^2 + bx + c$, $a \neq 0$.



- (i) Draw a neat labelled diagram to show the above situation diagrammatically. [1mark]
- (ii) If one zero of $(p - 1)x^2 + px + 1$ is 4, find the value of p. [2marks]
- (iii) Find the quadratic polynomial whose zeroes are 5 and -12 . [1mark]
- (iv) If one zero of $5x^2 + 13x + m$ is the reciprocal of the other, find m. [1mark]
12. Pooja was asked to find a polynomial whose zeroes are $3/\sqrt{2}$ and $-\sqrt{2}/5$. She calculated $\alpha + \beta = 13/(5\sqrt{2})$ and $\alpha\beta = -3/5$, and obtained $5\sqrt{2}x^2 - 13x - 3\sqrt{2}$. Her friend Neha claimed there was an error.
- (i) Is Neha correct in her claim? Give a reason. [1mark]
- (ii) If the polynomial is incorrect, find the correct one. If correct, verify by checking both zeroes. [2marks]
- (iii) Find the value of $\alpha^2 + \beta^2$. [1mark]
- (iv) If $(x + 3)$ is a factor of $q(x)$, find $q(-3)$. [1 mark]
13. Anjali is practising problems on fractions and pairs of linear equations.
- (i) A fraction becomes 2 when 2 is added to the numerator and 2 is subtracted from the denominator. Write the equation formed. [1mark]
- (ii) The same fraction becomes $1/3$ when only 2 is added to the denominator. Write the second equation. [1mark]
- (iii) Solve the two equations to find the original fraction. [2marks]
- (iv) For what value of k will $4x + y = 2$ and $(3k - 1)x + (k - 3)y = 3k + 5$ have no solution? [1mark]
14. A train covered a certain distance at a uniform speed. If the train had been 6 km/h faster, it would have taken 4 hours less than the scheduled time, and if the train were slower by 6 km/h, it would have taken 6 hours more than the scheduled time. Find the length of the journey. [5 marks]
15. The Chenab Railway Bridge in Jammu and Kashmir is the world's tallest railway bridge. Its shape is a parabolic arch whose equation is $ax^2 + bx + c = 0$. The nature of roots is given by $D = b^2 - 4ac$: $D > 0 \rightarrow$ real and distinct; $D = 0 \rightarrow$ real and equal; $D < 0 \rightarrow$ no real roots.



- (i) Find the nature of roots of $5x^2 - 4x - 3 = 0$. [1mark]
- (ii) Identify the type of roots of $x^2 + 3x + 3 = 0$. [1mark]
- (iii) Find k for which $3x^2 - 2x + 4k = 0$ has equal roots. [2marks]
- (iv) Find k for which $x^2 + 5kx + 16 = 0$ has no real roots. [1mark]



Answer key and step by step solution will be uploaded within 24 hours on App.